Sankalpa - Interactive Digital Learning Platform to Enrich Students with Special Needs

A.I Basnayaka
Dept of Computer Science and Software Engineering
Sri Lanka Institute Of Information Technology
Malabe, Sri Lanka
it20150266@my.sliit.lk

E.K.G.D. Epitakaduwa
Dept of Computer Science and Software Engineering
Sri Lanka Institute Of Information Technology
Malabe, Sri Lanka
it20142650@my.sliit.lk

Jayawardena C.P.U
Dept of Computer Science and Software Engineering
Sri Lanka Institute Of Information Technology
Malabe, Sri Lanka
it20228576@my.sliit.lk

Ashan Zoysa
Dept of Computer Science and Information Technology
Sri Lanka Institute Of Information Technology
Malabe, Sri Lanka
it20037024@my.sliit.lk

Uditha Dharmakeerthi
Dept of Computer Science and Network Engineering
Sri Lanka Institute Of Information Technology
Malabe, Sri Lanka
uditha.d@sliit.lk

Sanath Jayawardena
Dept of Computer Science and Network Engineering
Sri Lanka Institute Of Information Technology
Malabe, Sri Lanka
sanath.j@sliit.lk

2023 5th International Conference on Advancements in Computing (ICAC) | 979-8-3503-5813-1/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICAC60630.2023.10417380

***Abstract*— The term special needs mean the special educational needs that the students who suffer from learning disabilities must have due to several reasons. Learning impairment is a condition of the neurological system that impairs one's capacity to efficiently acquire and process knowledge. Failures in diagnosing the people with learning disabilities have been repeatedly highlighted and remain unpopular among people in the developing countries like Sir Lanka. Dyscalculia (difficulties in math), dysgraphia (difficulties in writing) and dyslexia (difficulties in reading) can be considered as some of the major learning disability issues. The 'mobile application based' solution "Sankalpa Math Master" has been created to offer a game based all-in-one solution including screening, and intervention to identify one of the above-mentioned learning disabilities called 'Dyscalculia'. Dyscalculia is generally distinguished by challenges in learning math, including organizing problems, multi-step calculations, transposing numbers, distinguishing directions, and understanding mathematical signs, leading to confusion with basic operations and facts. The provided solution has been accurate with the use of machine learning and deep learning techniques. In the process of screening with the use of Support Vector Machine (SVM) machine learning algorithm 99.77% of accuracy level has been obtained. The skill level prediction process has acquired an accuracy level of 97% with the use of Random Forest machine learning algorithm. Sri Lankan traditional education system will be vastly benefited by the high level of accuracy in screening, skill predicting, and other game and voice chat-based activities. The students with special needs will gain more effective and personalized learning experience which will improve their learning abilities and help to close the technology gap in education.**

Keywords*— *Dyscalculia, Probability identification, skill prediction, Image processing, voice enabled chat

I. INTRODUCTION

The concept of 'Special Educational Needs' or 'Special Needs' encompasses students with intellectual or developmental disabilities, and learning disabilities, as examples. When it comes to different parts of the world the term 'special educational needs' has been described in different ways. But in general, the term 'special education needs' is related to the students who are experiencing the learning difficulties that make them harder to engage in the normal learning process than the students who are in the same age [1]. Learning disability is a diverse set of neurological conditions that affect language processing, resulting in difficulties with reading, writing, math, and communication. Several types of learning disabilities exist in the world such as Dyscalculia, Dysgraphia, Dyslexia, etc. Referring to developing countries like Sri Lanka, the term 'learning disabilities' is not amiable with the people due to the lack of knowledge and traditional mind set.

Poor mathematics aptitude imposes an enormous cost on both society and affected individuals. This is called 'Dyscalculia' in medical terms [2]. This is one of the above-mentioned learning disabilities that have consistent difficulties using fundamental arithmetic procedures and memorizing math information. Dyscalculia is defined by difficulty processing numbers and amounts that often appear during years of preschool. Children with dyscalculia have difficulty with fundamental computations and face difficulties in dealing with more complicated problems in mathematics. Computation rules, deficits in math facts retrieval processes and Inability to switch from calculation to counting can be identified as such difficulties.

In the world there are a considerable number of children suffering with learning disabilities and remain undiagnosed due to deficiency in correct identification process or due to the technological gap. There are several existing applications that have been developed to screen dyscalculia. But none of them provides "all in one service" to these children. This problem heavily affects the developing countries like Sri Lanka and undeveloped countries such as Yemen. Children neither get a chance to enjoy the school life nor make a positive bond with studies [3] [4]. When it comes to Sri Lanka, students with learning disabilities can be mostly found themselves out of the main education stream resulted

979-8-3503-5813-1/23/$31.00 ©2023 IEEE

Authorized licensed use limited to: University of New South Wales. Downloaded on October 26,2025 at 02:29:51 UTC from IEEE Xplore. Restrictions apply.

in escalation of high school ‘drop-out’ rate. Within traditional society, especially in the countries like south Asia, parents and school authority pays a high attention on talented students over weak students. They even do not believe the technology as a remedy for the educational weakness of a child. As per United Nations International Children's Emergency Fund (UNICEF) calculations, 23.5% of children within the age limit of 5-14 years are excluded from the normal educational stream and reduce participation in the general educational activities [5].

Within this research school students from grade 1 to 8 have been taken as the sample to discuss the above-mentioned learning disability. Main objective of this research is to carry out a solution to help the school students with a self-study friendly application solution, including screening, intervention, and a skill prediction process for dyscalculia. The intervention includes ‘finger math concept’ [6] based activities and ‘voice chat bot’ [7] based self-support system. During the prediction, guardians can access the application and get the chance to track the development of the children. This has been planned to deliver among school students in both primary and secondary classes.

II. Literature Review

Learning disabilities are a neurological disorder hindering information processing, without known cures or therapies. Limited awareness among parents and instructors in Sri Lanka leads to undiagnosed cases. International research groups aim to develop tools for dyscalculia detection in children, yielding various solutions through their efforts. Often, solutions aim to identify multiple disabilities, but lack crucial functionalities. Some require supervision, making accurate diagnosis and treatment challenging. Users face difficulty in diagnosing conditions and require multiple solutions for comprehensive treatment and predictions.

The “Dyscalculia Screener” [8] has been developed by Brian Butterworth with the aim of identifying the students between age 6-14, who are suffering with learning disabilities in mathematics. He had used several kinds of quizzes including dot enumeration, Number Comparison (also referred to as Numerical Stroop) and Arithmetic Achievement test which only contains addition and multiplication related questions. This application is a low interactive application used in desktop computers.

” TALASIA” mobile application by Meister Cody is another approach [9]. This was designed by Prof. Dr. Jörg-Tobias Kuhn and Dr. Christin Schwenk. They have provided a test to identify the dyscalculic students and have also provided several training sessions to improve number skills. The activities in this application are mainly based on a land called ‘TALSIA’ and around a magician and dragons in that country. This focuses the elementary students with an age limit of 2 – 4.

But when studying the process of both above applications there were few similar drawbacks such as not using any voice commands to support children, not using IQ test to accurate the result and no skill prediction service for students.

Fingermath – Arithmetic Hand Gesture Game is an Interactive game-based approach for early childhood education [10]. This has been developed especially with the purpose of teaching counting and basic mathematics skills like addition and collection. The innovative 'jarimajic' method provides an engaging and effective way for young learners to grasp these concepts. Use hand images to do calculation games. But a few downsides of this can be addressed as limited multiplications and low interactive approach in using hand gestures.

"Nana Shilpa" is a mobile app designed to assist in screening and addressing various learning difficulties [11]. It focuses on verbal dyscalculia, lexical dyscalculia, operational dyscalculia, pragmatic dyscalculia, dysgraphia, and numeric dysgraphia. They also provide several small games and activities to train the skills of the students. The app utilizes deep learning methods for more effective screening. But as a downside it does not use Intelligence Quotient (IQ) level testing, student skill prediction or finger math concepts to get more accuracy to the project.

‘MathFun: A Mobile App for Dyscalculia Children’ is research done by Rizawati Rohizan, Lim Hean Soon and Dr. Siti Azreena Mubin [12]. This was a mobile application built to help dyscalculic students in learning and engaging in math. It contains learning and exercise processes. They focus on helping students to familiarize themselves with mathematical symbols. Development has been done according to the ‘Calculic Model’. But this approach lacks the Dyscalculia detection process, skill prediction process or finger math concepts.

Mohamad Firdaus Che Abdul Rani, Rizawati Rohizan, and Nor Azlina Abd Rahman developed a web-based solution for primary students with dyscalculia and dyslexia [13]. The solution provides specialized activities and game-based interactions to enhance their knowledge in an engaging manner. There are several counting and writing activities provided. This is mainly based on helping the students with activities. It does not focus on the screening of the disability.

‘Mobile application for dyscalculia children in Malaysia’ which has been developed by Mazeyanti Mohd Ariffin, Fiqa Azureen Abd Halim, and Norshakirah Abd Aziz [14]is an application that is focusing on the teaching concepts for the dyscalculic students. This application mainly focused on counting numbers, counting objects, subtraction, and addition related operations. This is a much more improved application as this has been developed based on the characteristics of dyscalculic students. The activity in this application is based on finger counting and motivational sounds like clapping sounds are also included.

Both above solutions are based on only activity-based gaming environment. These solutions lack some functions like dyscalculia screening, skill prediction, a voice enabled chatting system and fingermath concepts.

Development and Standardization of Dyscalculia Screening Tool (DST) [15] is the latest dyscalculia screening approach that has been created to screen the dyscalculia disability. In this approach the participant has to answer several questions by simply selecting the ‘Yes’ and ‘No’ buttons. Selecting ‘Yes’ button adds 0 mark to your total mark and selecting ‘No’ adds 1 mark to your total mark. The result will be given by the participant based on the total marks obtained. This application only focusses on several predesigned questions but does not focus on the participants’ current skills and IQ knowledge.

Authorized licensed use limited to: University of New South Wales. Downloaded on October 26,2025 at 02:29:51 UTC from IEEE Xplore. Restrictions apply.

After clearly analyzing all the drawbacks of the existing approaches, 'Sankalpa' Math Master has been proposed. This new approach provides a more accurate and more interactive experience for the students using new technologies. Simple and interactive screening process, voice enabled chat bot as a digital instructor, fingermath concept to train the skills and skill prediction process has been supplied in this new approach for this learning disability.

III. Methodology

The "Sankalpa" is a learning platform for students who are suffering from dyscalculia. This mobile application provides some features such as identifying dyscalculia probability of the student, Voice enables chat bot, Arithmetic learning system using finger math concepts and provide performance analysis skill level predication. Fig 1 illustrates the system overview of the "Sankalpa" mobile application.

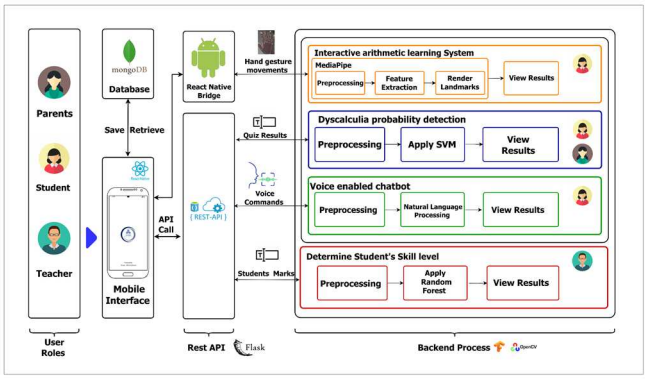


Fig. 1. System Overview Diagram

A. *Identify the probability of having dyscalculia and determine the study focus area*

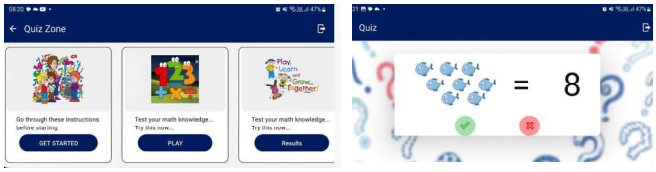


Fig. 2. Dyscalculia identifying tests component interfaces.

In the Probability identification, several steps were carried out to do the screening process to identify the low or high probability of having dyscalculia among school students. To start with, an IQ test was given as shown in the Figure 1, to check the IQ level of the student. Meanwhile, a small quiz with a few questions was also raised for the guardian to get the data of the student in his/her point of view. Finally, an interactive mathematical quiz has been carried out to identify the probability of having dyscalculia.

Moving on to the model training process, 551 quiz results have been collected from students who are both dyscalculic and non-dyscalculic. The students belong to both primary and secondary level classes in school. If consider the age, they all belong to the age group between 6-13 years. The model was trained using python language using Jupiter lab. Scikit-learn (Sklearn) library was used to create the model. For the mathematical quiz 2 sets of separate quizzes have been created according to age groups. Only 10 quizzes will be given in one attempt for a one student to face. Each quiz gets 30 seconds to give the answer and if a student is unable to answer the question will be skipped with zero marks. MongoDB has been used as the database.

In this context, the process involves performing classification by utilizing a Support Vector Machine (SVM) supervised learning algorithm. Eemploying the SVM algorithm is aimed at locating the hyper-plane in order to accurately classify future data points. This SVM algorithm helps to predict whether the student has a low or high probability of having dyscalculia. To get more accurate models using a non-linear dataset, the polynomial kernel has been used in the training process.

For the probability identification process, the marks of the IQ test that was answered by the student, marks of the questionnaire answered by the parent or the guardian, and marks of the mathematical quiz performed by the student, have been used. All these features have been fed into the SVM model. This helps to get the output of the SVM model to identify the students low or high probability level of having dyscalculia. The student may see the marks they have got for several questionnaires and the final prediction result at the end of the process in a separate page as a summary report.

If the student has a high probability of having dyscalculia, they will be informed of the facilities provided within our application to help them. They will be given a guide of how to use the in-app facilities for their improvement. After getting the results and guidance, students can head on to the dashboard of the application to start using the other facilities. Students can use this facility again if they need to see whether they are cured or not. After more medical professional consultations, the decision has been made to elevate the guideline providing facility to a higher level.

B. *Developing a voice-enabled chat bot to assist children with dyscalculia..*

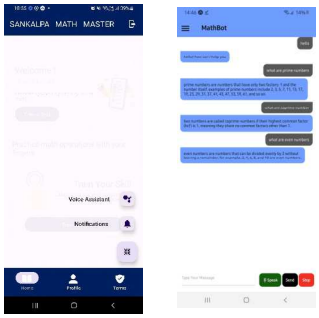


Fig. 3. Chatbot component interfaces.

This voice-enabled chatbot has been designed for children to offer an interactive and engaging arithmetic learning experience. The chatbot communicates amiably, converting voice inputs to text for on-screen interaction. It analyzes children's questions, corrects errors, and provides feedback for better understanding. By suggesting answers and displaying results, the chatbot acts as a supportive virtual instructor, enhancing math comprehension, problem-solving skills, and fostering a positive learning atmosphere for children. [16]

This employs Python's speech recognition for its speech-to-text capability. It effectively translates spoken words into text, facilitating seamless user-chatbot interaction.

Authorized licensed use limited to: University of New South Wales. Downloaded on October 26,2025 at 02:29:51 UTC from IEEE Xplore. Restrictions apply.

Utilizing the library, the chatbot processes verbal input captured through the device's microphone. This versatile library excels in converting audio into corresponding text, accommodating various sources like microphones or audio files. Support of it for the multiple voice recognition engines and Application Programming Interface (API)s allows developers to choose the most suitable one, ensuring accurate language support and compatibility across different operating systems.

Once the voice input has been captured and translated to text, it can continue processing and analyzing the text-based input to extract necessary information. The 'Multinomial Naive Bayes algorithm' is used for Natural Language Processing (NLP) in the development of the voice-enabled chatbot. The model has been built using a dataset consisting of math-related questions along with their corresponding answers. With the use of this dataset, the chatbot can be trained to comprehend user queries and provide the most accurate responses. The Multinomial Naive Bayes algorithm was selected for its efficiency when dealing with text classification problems, which fits well with the intention of the chatbot, which is to comprehend queries from users and provide appropriate responses. Based on the Bayes theorem, this approach assumes that characteristics are conditionally independent given the class. It determines the probability that a given text belongs to a specific class and chooses the class with the highest probability to be the predicted class. The chatbot learns from the dataset during the training phase by estimating the Multinomial Naive Bayes model's parameters. To comprehend the patterns and connections between various concepts in mathematics and related queries, it analyzes the question-answer pairs. This procedure helps the chatbot to determine the purpose of user inquiries and obtain pertinent information from its knowledge base. After the model has been trained, the chatbot could apply the Multinomial Naive Bayes algorithm to assess user queries. For the provided question, it determines the probabilities of different classes (such as math topics) and chooses the most likely class. The chatbot searches its knowledge base for the appropriate response based on its predicted class and responds to the user with it.

C. Develop interactive arithmetic learning system

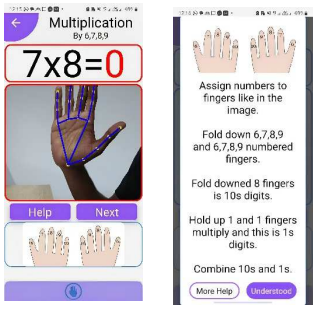


Fig. 4. Finger maths learning component interfaces.

Integrating Image Processing Technology into childhood lessons fosters a dynamic learning environment, tailoring education to individual needs and enhancing engagement. This approach accelerates learning and produces educational activities utilizing image processing software for basic math problem-solving using fingers. The initiative aims to create a reliable classroom tool and contribute to understanding image processing's effectiveness as a teaching aid for young children, enriching educational practices and knowledge.

For example, one way may entail employing cameras to capture photos of children's fingers while they answer math problems. The Image Processing Technology may then evaluate the photos to determine whether the calculations are right. This not only teaches numerical skills but also helps youngsters to learn and build finger coordination [17].

The use of image processing technology to solve simple math concepts using hand gestures and finger counting is an underutilized solution in online learning.

Utilizing finger math techniques to create lessons on addition, subtraction, and basic multiplication (6 to 10) with interactive live practical tests. Doing that can elevate children's effectiveness and efficiency. And they can easily understand arithmetic concepts and practice arithmetic problems on their own. Make the learning experience more interactive and fun for children. Children can actively participate in the learning process through hand gestures and finger counting. This provides them with immediate feedback on their understanding of math concepts and helps to increase their engagement and motivation. It has the potential to enhance the learning experience for children, especially in the early childhood years, by making it more interactive, engaging, and effective.

First children can access the lessons and refer to how to solve arithmetic problems using finger-math techniques and they can interact with the device using their hands and fingers to solve arithmetic problems that they learned in the lessons and practice and get the feedback live. The system will generate different types of arithmetic problems automatically and constantly. When children struggle to give the correct answer, the system will help to solve that problem by giving feedback to the children.

To develop a Finger count identification and display finger overlay system first tried using open-cv with a custom dataset. However, after developing a system using Convolutional Neural Network (CNN) architecture and many preprocessing techniques, the system struggled a lot. Need perfect lighting conditions, it is hard to identify both hands and hard to display the finger point overlay. Performance issues, Lack of accuracy.

This system should be very fluid and accurate because this system has been designed for small children with Dyscalculia. And especially this is a mobile application therefore it needs to be very efficient and run on low-end mobiles. The MediaPipe Library developed by Google has been found when searching for different types of technologies to solve these problems. The library also supports displaying finger landmarks overlay on top of the camera feed, which is a significant advantage for this system as it enhances interaction and greatly benefits small children. The Media pipe library can identify both hands at the same time and detect 21 landmarks pointing in a single hand in a single frame. By employing that library and custom finger landmark point values designed for children, the system is refined to accurately recognize the finger count on both the left and right hands. And the system can identify the fingers

Authorized licensed use limited to: University of New South Wales. Downloaded on October 26,2025 at 02:29:51 UTC from IEEE Xplore. Restrictions apply.

respectively as their name. Using those details, the system can identify fingermath techniques and the answers to the system-generated test question and help the student to interact and learn basic mathematics in a more relaxed fun way.

D. *Dyscalculia Student's Academic Performance Skill level predication*

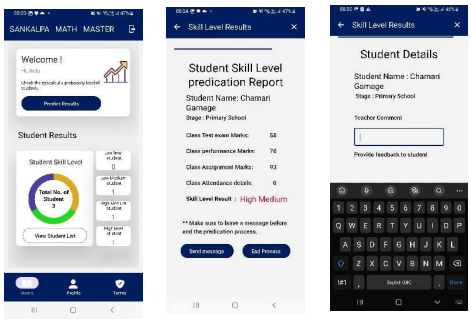


Fig. 5. Teacher Dashboard and Student's skill level predication component interfaces

The proposed system called "Sankalpa" provides the ability to predicate the skill level of the dyscalculia student and more other features to enhance the quality of education for students who suffer from dyscalculia. Teachers can use the skill prediction process to input the assignment marks of their students and other details. The marks have been collected and analyzed to assess the mathematical abilities of the students and identify any challenges they may face due to dyscalculia. Parameters such as class test marks, assignment marks, and overall performance in mathematical assignments are considered. This data has been subsequently used to predict the skill level and academic performance of each student.

To develop the prediction model, the application utilizes Machine Learning (ML) techniques. The model has been trained using a dataset that includes information on the performance of dyscalculia students in their class assignments. Based on this training data, the ML model can analyze the assignment marks provided by the teacher and provide a level indicating each student's skill level in mathematics and their potential academic performance. Jupyter notebook has been used to implement this process. It is open-source software, open standards, and services for interactive computing across multiple programming languages. The data has been organized using this tool. Then supervised neural network has been designed to train the data and extract information from the data. In our experimental case, we predict the student's performance. In this system, student performance has been predicted, and the accuracy percentage has been determined. Once the prediction process has been completed, the application presents the results, including the predicted skill level, academic performance score, and any additional insights or recommendations. Teachers can utilize this information to understand their students' mathematical abilities and plan appropriate teaching strategies.

If the prediction suggests that a student may have difficulties or a lower skill level in mathematics, the application provides intervention options that are accessible to teachers. These interventions may involve personalized study plans, additional practice exercises, or recommended educational resources designed to address the specific needs of each student.

Additionally, the application allows teachers to access comprehensive reports that include the results of the Skill prediction process and could notify the feedback personally to the student profiles. This feature enables teachers to monitor the development of their students over time and make informed decisions about ongoing interventions and academic support.

The Sankalpa application provides a teacher-focused feature that allows for the input and analysis of class assignment marks to predict the skill level and academic performance of dyscalculia students. It offers targeted interventions and support, empowering teachers to guide their students toward improved mathematical abilities and academic success.

IV. Results and Discussions.

A. *Identify the probability of having dyscalculia and determine the study focus area*

The SVM model was fed with the data that were collected from schools and therapy centers. The above-mentioned features were used to train the SVM classification algorithm. 99.77% accuracy could be achieved by the algorithm for the probability detection. The test accuracy was shown nearly 98.5%.

B. *Developing a voice-enable chat bot to assist childern with dyscalculia.*

The Multinomial Naive Bayes algorithm has been used as the algorithm to proceed with the voice enabled chat bot. The model has been trained using this algorithm and this uses a theorem called Bayes theorem to proceed. The data set has been fed into the algorithm and trained to get the best accuracy in test and result data.

C. *Develop intractive arithmatic learning system*

Media Pipe Hand is an exceptional machine learning-based library that stands out with its real-time hand detection and tracking capabilities, providing precise 21 2D hand landmarks. It's ensuring real-time performance on mobile devices. With an impressive accuracy of 95.7% for palm detection and 91% for hand landmark detection, it can reliably handle occlusions and lighting variations, delivering consistent and robust results. Its unique ability to detect and track multiple hands simultaneously makes it particularly valuable for complex hand interactions and group scenarios.

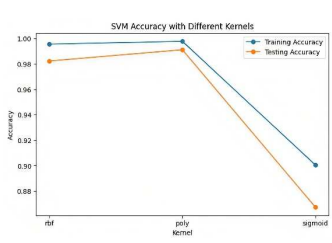
D. *Progress analysis and Skill performance level predication*

The Random Forest model was utilized to analyze a dataset consisting of 1024 samples. This dataset was employed as training data for the Random Forest classification algorithm. The model was trained using five specific features to predicate children's skill level in four development phases. The results demonstrated a remarkable accuracy rate of approximately 98% in skill predication.

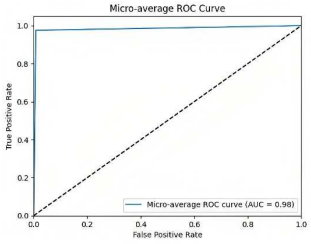
To evaluate the model's performance further, 1/3 of instances were selected for testing purposes. Impressively, the Random Forest model correctly classified testing accuracy rate of 97%. Only one instance was incorrectly

Authorized licensed use limited to: University of New South Wales. Downloaded on October 26,2025 at 02:29:51 UTC from IEEE Xplore. Restrictions apply.

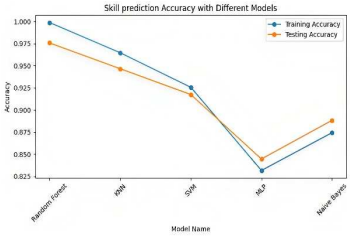
predicted, highlighting the model's effectiveness in accurately predicting the skill level of the student.



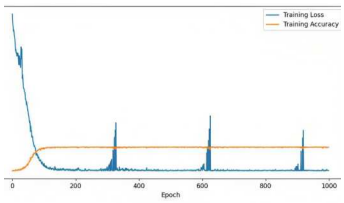
(b) SVM model accuracy for Dyscalculia identification



(a) ROC curve for Skill level predication



(d) Random forest model accuracy for Skill predication



(c) Accuracy graph for voice enable chat-bot.

Fig. 6. Accuracy evidence

V. Conclusion

There were limited research projects conducted in the field of learning disabilities, particularly in fulfilling the requirements mentioned. However, an innovative mobile-based solution called "Sankalpa" has been developed to address this gap. This solution offers various features, including the ability to identify the probability of dyscalculia in students, a voice-enabled chatbot, an arithmetic learning system using finger math concepts, and Skill level predication for dyscalculia school students.

Currently, the "Sankalpa" application is available on the Android platform and supports the native language, Sinhala. However, there are plans to expand its reach in the future. The goal is to make the solution accessible in multiple languages, including support for platforms like iOS, and even provide a web-based version. By pursuing these enhancements, the aim is to develop a comprehensive and improved solution that fulfills the aforementioned requirement.

Acknowledgment

The researchers extend their sincere appreciation to Dr. Ranjith Siriwardena, Consultant Psychiatrist at Provincial General Hospital, for his invaluable guidance in the field of medicine. Gratitude is also extended to the principals, teachers, students, and parents of MR/Sapugoda Sir Rathanapala School for their unwavering support and invaluable contributions throughout the research. The group members are recognized for their remarkable dedication from project inception to completion.

References

[1] Kohli, Adarsh and Sharma, Samita and Padhy, Susanta, "Specific Learning Disabilities: Issues that Remain Unanswered," *Indian Journal of Psychological Medicine,* p. 40, 2018/09.

[2] Andrea Battisti, Giulia Lazzaro, Cristiana Varuzza , Stefano Vicari and Deny Menghini, "Distance learning during COVID-19 lockdown: Impact on adolescents with specific learning disorders and their parents," *Frontiers in Psychiatry,* vol. 13, 2022.

[3] Karunanayake, Danesh and Dilrukshi, Chameera and N.D, Umesha, "How Teaching Styles Affect to Enhance Students' Mathematical Education," *International Journal of Scientific Research in Science and Technology,* vol. 7, 2020/11.

[4] Shalini Felicity Wikremesooriya, "Teaching Children with Learning Diiculties via Community-Based Rehabilitation Projects in rural Sri Lanka," *Disability, CBR, and Inclusive Development Journal ,* vol. 26, p. 7, 2015.

[5] "EVERY MIND".*www.unicef.org.*

[6] R. Greenberg, "Image Processing for Teaching," *Journal of Science Education and Technology,,* p. 2, 1993/09.

[7] B. Butterworth, Dyscalculia screener, NferNelson Pub., 2003.

[8] M. Schütze, "Talasia (Dyscalculia) – Meister Cody," Meister Cody GmbH,[Online].Available: www.meistercody.com/educationalresearch/talasia-math/?lang=en.

[9] Yuliana, Dewi and Suharjito, "Fingermath – Arithmetic Hand Gesture Game to Improve Early Childhood Mathematics Learning," *2019 International Conference on Information Management and Technology(ICIMTech),*vol.1,no. 10.1109/ICIMTech.2019.8843756, pp. 6,11, 2019.

[10] Hewapathirana, Chamil and Abeysinghe, Kalpani and Maheshani, Pamodya and Liyanage, Prabath and Krishara, Jenny and Thelijjagoda, Samantha, "A Mobile-Based Screening and Refinement System to Identify the Risk of Dyscalculia and Dysgraphia Learning Disabilities in Primary School Students," *2021 10th International Conference on Information and Automation for Sustainability (ICIAfS),* pp. 287-292, 2021.

[11] Rohizan, Rizawati and Soon, Lim and Mubin, Siti Azreena, "MathFun: A Mobile App For Dyscalculia Children," *Journal of Physics: Conference Series,* vol. 1712, 2020/12.

[12] Che Abdul Rani, Mohamad Firdaus and Rohizan, Rizawati and Abd Rahman, Nor, "Web-based learning tool for primary school student with dyscalculia," pp. 157-162, 2014.

[13] Ariffin, Mazeyanti and Abd Halim, Fiqa Azureen and Aziz, NorShakirah, "Mobile Application for Dyscalculia Children in Malaysia," 2017/04.

[14] Jeya, Adaikala and Pio Albina, A, "{Elementary Education Online," *Development and Standardization of Dyscalculia Screening Tool (DST),* vol. 20, pp. 1982--1982, 2021.

[15] K. B. Eashwar, R. Venkatesan,D. Ganesh, "Student performance prediction using SVM," *International Journal of Mechanical Engineering and Technology (IJMET),* vol. 7, p. 649–662, 2017/11.

[16] Mason, John, Lerman, Stephen, Questioning in Mathematics Education, Springer Netherlands, 2014.

[17] Yanuarto, Wanda, "How a brain says: Fingermath for Empowering Children's Creativity," *Journal of Education and Learning (EduLearn),* vol. 11, p. 404, 2017/11.

Authorized licensed use limited to: University of New South Wales. Downloaded on October 26,2025 at 02:29:51 UTC from IEEE Xplore. Restrictions apply.